Travaya — AI Chatbot Module (Triva)

Module: AI & Recommendation System

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1. File Structure

```
travaya_chatbot/
|
|-- chatbot_api.py          <- Flask API server (run this)
|-- intents.json            <- Model 1: General travel intents
|-- intents1.json           <- Model 2: India destination-specific
   details
|-- TrivaChatbot.jsx        <- React component for MERN frontend
|
|-- data.pickle             <- Auto-generated: preprocessed Model 1 data
|-- data1.pickle            <- Auto-generated: preprocessed Model 2 data
|-- model_weights.h5        <- Auto-generated: trained Model 1 weights
|-- model1_weights.h5       <- Auto-generated: trained Model 2 weights
|
\-- README.md               <- Documentation
```

2. How to Run

2.1. Step 1: Install Dependencies

```
pip install flask flask-cors tensorflow nltk
```

2.2. Step 2: Start the Flask Chatbot API

```
python chatbot_api.py
```

2.3. Step 3: Integrate with React (MERN Frontend)

Copy `TrivaChatbot.jsx` into your React project's components folder, then add it anywhere in your app:

```
// In App.jsx or any layout component
import TrivaChatbot from './components/TrivaChatbot';

function App() {
  return (
    <div>
      {/* Your existing MERN app */}
      <TrivaChatbot />   {/* Add this one line */}
    </div>
  );
}
```

Listing 1: App.jsx — Adding Triva to MERN frontend

The chatbot will appear as a floating button (bottom-right corner) on every page of your website.

3. How the Chatbot Works

3.1. Two-Model Architecture

Model 1 — General Intent Classifier

- File: `intents.json`
- Handles: greetings, goodbyes, general package info, booking info, travel styles
- Tags: `greeting`, `goodbye`, `identity`, `destinations`, `budget_travel`, `adventure_travel`, etc.

Model 2 — Destination-Specific Responder

- File: `intents1.json`
- Handles: specific Indian state/destination details with full package info
- Tags: `Goa`, `Kerala`, `Rajasthan`, `Kashmir`, `Andaman`, `Himachal`, `Ladakh`, etc.

3.2. Routing Flow

```
User Message
      |
      v
Model 1  -->  Classifies intent (tag)
      |
      v
If tag == "destinations"  -->  Model 2  -->  Returns specific package
   details
Else                        -->  Model 1  -->  Returns general response
```

4. Covered Indian Destinations

Destination	Package Cost	Duration	Type
Goa	INR 12,000 – INR 18,000	5D/4N	Beach, Water Sports
Andaman & Nicobar	INR 28,000 – INR 45,000	7D/6N	Beach, Scuba Diving
Kerala	INR 20,000 – INR 32,000	7D/6N	Backwaters, Nature
Rajasthan	INR 20,000 – INR 38,000	10D/9N	Heritage, Culture
Kashmir	INR 25,000 – INR 40,000	6D/5N	Nature, Adventure

Destination	Package Cost	Duration	Type
Himachal Pradesh	INR 22,000 – INR 35,000	8D/7N	Adventure, Trekking
Ladakh	INR 25,000 – INR 42,000	7D/6N	Adventure, Biking
Darjeeling & Sikkim	INR 18,000 – INR 28,000	7D/6N	Nature, Culture
Varanasi & Agra	INR 12,000 – INR 20,000	5D/4N	Spiritual, Heritage
Northeast (Meghalaya/Assam)	INR 22,000 – INR 36,000	8D/7N	Nature, Wildlife
Karnataka	INR 14,000 – INR 24,000	6D/5N	Heritage, Nature
Tamil Nadu	INR 15,000 – INR 26,000	7D/6N	Culture, Spiritual
Maharashtra	INR 14,000 – INR 22,000	6D/5N	Heritage, City
Gujarat	INR 16,000 – INR 28,000	7D/6N	Wildlife, Heritage
Lakshadweep	INR 30,000 – INR 55,000	5D/4N	Luxury, Beach
Odisha	INR 11,000 – INR 18,000	5D/4N	Spiritual, Beach

5. API Reference

5.1. POST /api/chat

```
// Request Body
{ "message": "Tell me about Kerala" }

// Response
{
  "status":     "success",
  "response":   "Kerala - God's Own Country\n- Duration: 7 Days...",
  "tag":        "destinations",
  "confidence": 0.94
}
```

Listing 2: Request and Response — /api/chat

5.2. GET /api/health

Returns server status and a list of all supported destinations.

5.3. GET /api/destinations

Returns all destination tags currently supported by the chatbot.

6. Technology Stack

Component	Technology
NLP Engine	TensorFlow 2.x + Keras
Text Processing	NLTK + Lancaster Stemmer
API Server	Flask + Flask-CORS
Frontend Component	React.js (JSX)
Data Format	JSON (intents) + Pickle (preprocessed)
Algorithm	Bag-of-Words + Dense Neural Network

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